WaterBuddy planning and design

A clear blueprint for the matching Python and Streamlit app in FA2

THE USER PROBLEM

People can forget to drink while studying or working, and logging every amount can feel like extra effort. WaterBuddy makes the next action easy to find and shows immediate, understandable feedback.

Simple enough for a short break

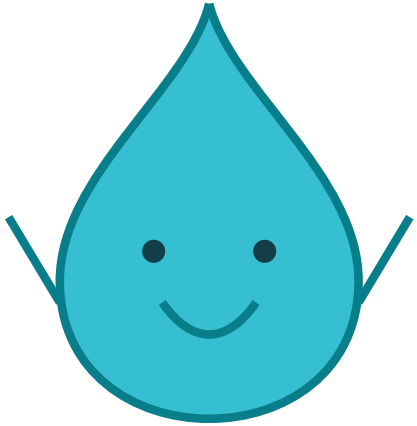
A visible +250 ml button, optional container fractions and a custom field reduce repeated typing.

Encouragement with a clear stopping point

A friendly drop reacts at 50%, 75% and 100%. Points stop increasing after completion; there is no drinking leaderboard.

Privacy and honest expectations

No account or personal identifier. Records exist in the current server session; downloads preserve a readable personal record.



DESIGN OBJECTIVE

Choose a goal, log actual water, understand progress and build a manageable routine.

Research approach

Desk research and brief analysis, not a user interview study. Forgetfulness and app friction are design hypotheses from the brief; no survey results or participant quotations are invented.

Age aware goals and first use

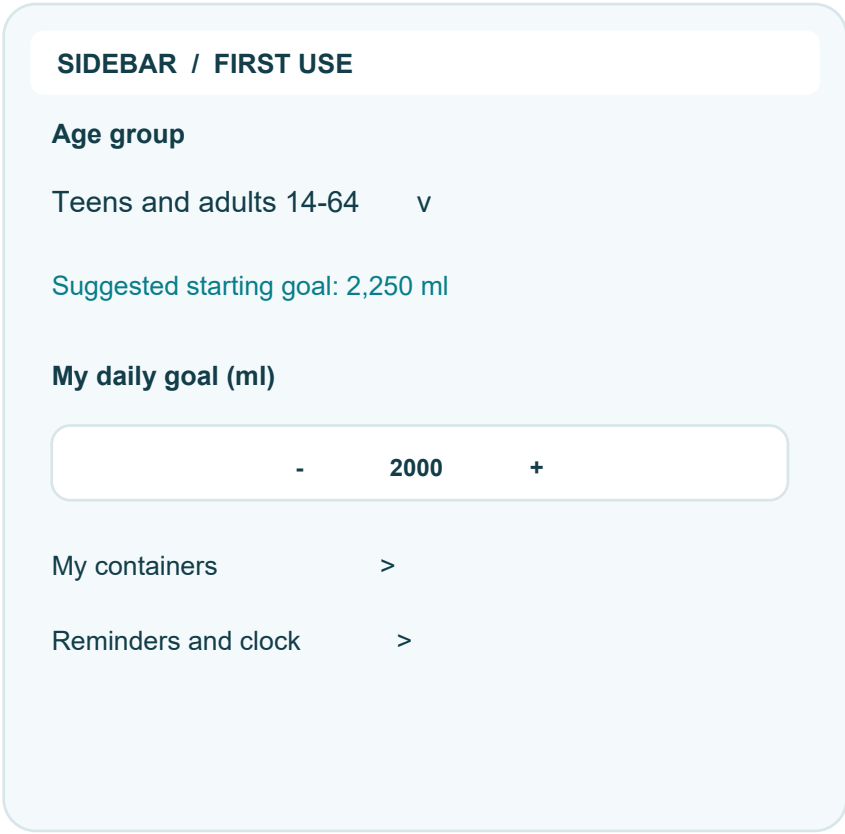
Settings remain in the sidebar so users can change their goal without losing water already logged

ILLUSTRATIVE LOGGING PRESETS

Age group	Starting goal
Children 4-8	1,200 ml
Young people 9-13	1,700 ml
Teens and adults 14-64	2,250 ml
Older adults 65+	1,800 ml

These are editable classroom presets carried forward from the project concept. They are not universal drinking-water recommendations. The app labels them as starting goals.

The About page separately explains EFSA total-water references, which include food and all drinks. A user logging plain water should not confuse these two quantities. [1]



Action: select an age group. Python looks up its preset. Adjust the goal from 250 to 5,000 ml; existing logs stay intact.

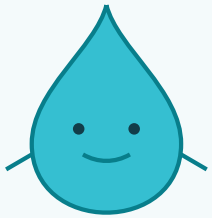
Age-aware support

Children choose an appropriate goal with an adult. Individual clinical guidance takes priority over any app preset.

Home and milestone reactions

Four states of the same dashboard | Illustration uses an adjusted 2,000 ml goal

HOME / 0%



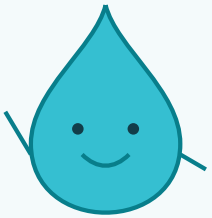
0 / 2,000 ml

2,000 ml remaining

Ready when you are

No logs yet. Total = 0; progress = 0%; buddy is ready.

HOME / 50%



1,000 / 2,000 ml

1,000 ml remaining

Halfway there

At 50%, the drop smiles and waves. Remaining = 1,000 ml.

HOME / 75%



1,500 / 2,000 ml

500 ml remaining

Your buddy is cheering

At 75%, both arms rise. The message changes immediately.

HOME / 100%



2,000 / 2,000 ml

0 ml remaining

Goal complete

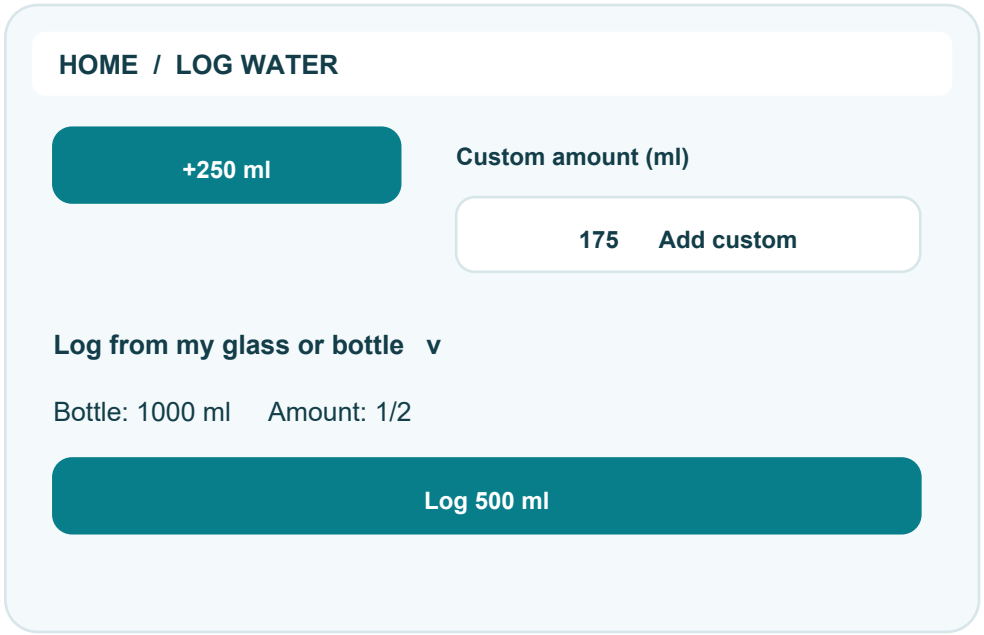
At 100%, celebrate and unlock Hydration Hero. Daily rewards are capped.

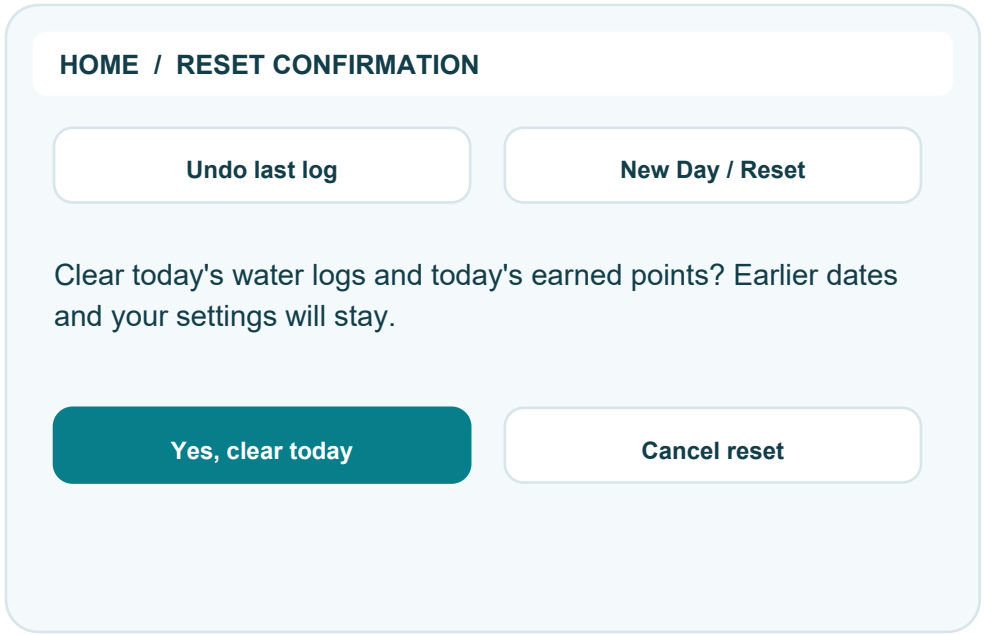
ONE CONSISTENT DASHBOARD

Home shows total, remaining, chosen goal, true percentage, progress bar and a comparison with the age preset. At 2,250 / 2,000 ml, show 112.5%, remaining 0 ml and a full bar.

Logging and a safe reset

Controls are sections within Home, matching the final app rather than separate setup screens





1 Quick log appends 250 ml. Custom input accepts whole values from 1 to 2,000 ml. Invalid values cannot enter the log.

2 Container estimate = size x fraction. A 1,000 ml bottle x 1/2 adds 500 ml. Fractions are 1/4, 1/2, 3/4 and full.

3 Undo removes only the last drink. All totals, visuals and points recalculate from the remaining log.

4 Cancel keeps the data. Confirm clears today only. A reset does not create a new date or increase a streak.

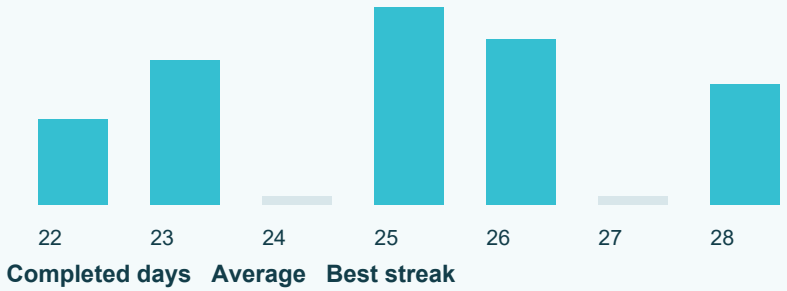
Calendar rollover

The next actual local date starts a fresh record automatically on interaction, or on the 60 second Home check. Earlier session records stay available.

Progress and rewards

These are planned examples, not screenshots of claimed personal history

PROGRESS / LAST SEVEN DAYS



Download weekly CSV

Session backup JSON

REWARDS / DAILY ACHIEVEMENTS

AquaPoints	Calendar streak
Up to 125 today	Consecutive real dates

50% Halfway Hero

75% Steady Sipper

100% Hydration Hero

At 100% choose a celebration accessory

A dated chart, status table and today's drink log explain the numbers. Missing dates say No record; the average uses recorded dates only.

Points = min(100, floor(total / goal x 100)) + 25 on completion. A total above the goal earns no extra points.

History belongs to this session. JSON and CSV are readable exports; automatic backup restore and permanent history are outside this version.

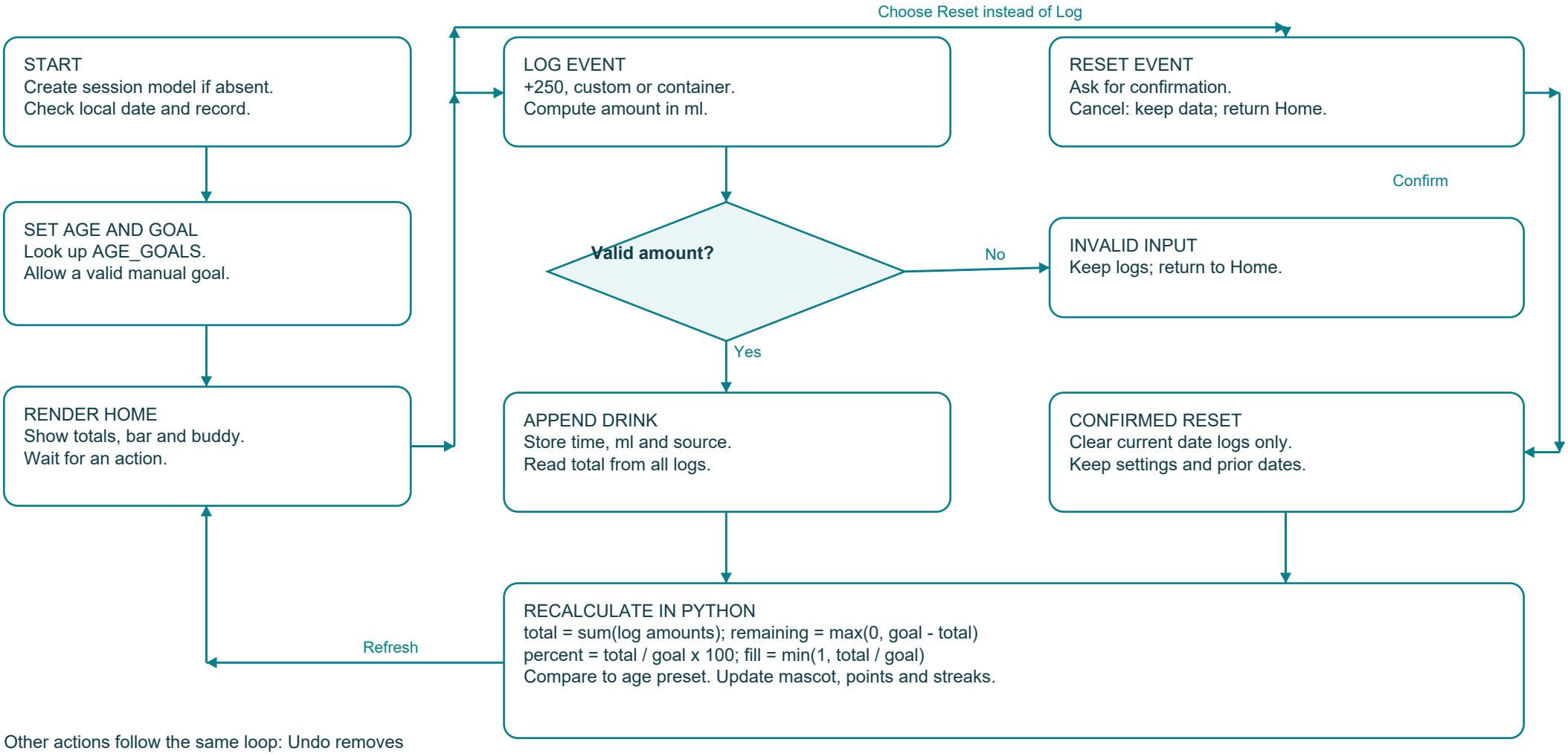
A streak counts saved, consecutive calendar dates meeting each day's goal. Goal changes, undo and reset recalculate points and badges.

Optional features support the core task

Mini missions use 25%, 50%, 75% and 100% checkpoints. Celebration accessories are cosmetic; there are no promised notifications or rewards outside the app.

Main Python logic and event flow

Follow the arrows from initialization to an action, then back to the refreshed interface



Other actions follow the same loop: Undo removes the final log; age or goal changes update the goal and preserve intake. [Python]

Feedback decisions and reminders

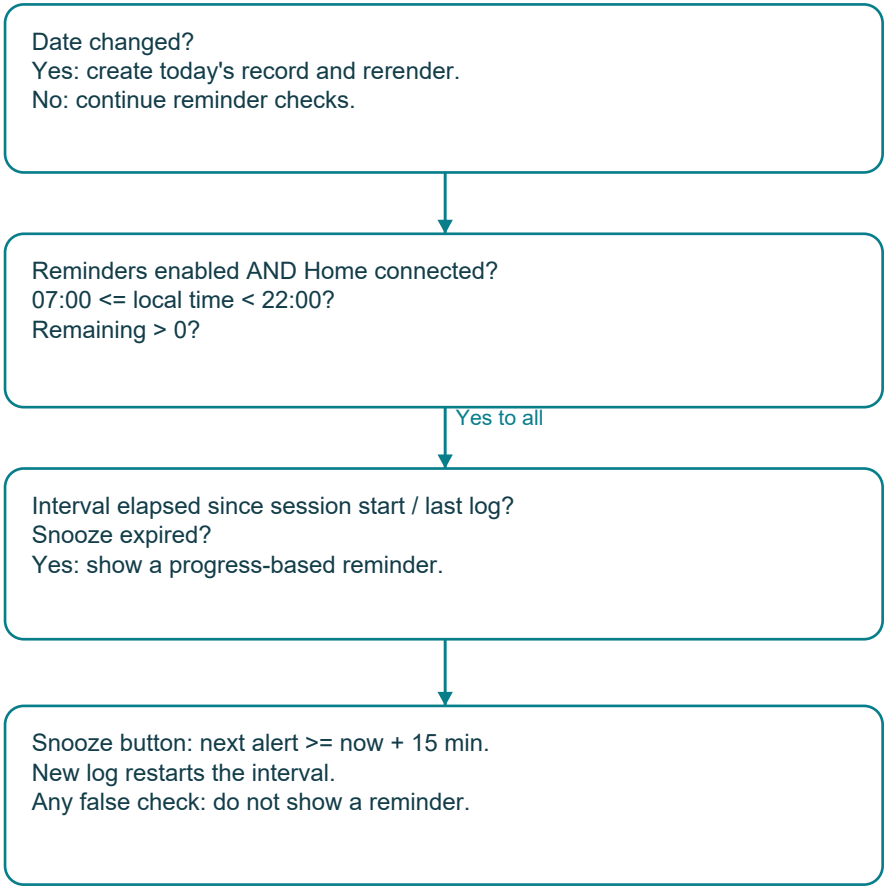
Detailed rules prevent stale totals, duplicate rewards and misleading notifications

MILESTONE PRIORITY

Check in this order	Buddy reaction	Output
percent >= 100	Complete	Goal complete and full bar
else percent >= 75	Cheer	75% encouragement
else percent >= 50	Wave	Halfway message
else percent > 0	Smile	Good start
else	Ready	Log your first drink

Check the highest threshold first, so a completed goal never shows a halfway message. Visual fill is capped but the numerical percentage is accurate above 100%.

ON EACH 60 SECOND HOME CHECK



State stored in st.session_state

model = age + chosen goal + current date + dated logs.
Each drink stores ml, time and source.
The interface uses Streamlit widgets; SVG replaces desktop Turtle so mascot reactions work in the deployed web app.

From storyboard to the FA2 app

Every planned control has a direct implementation and a checkable outcome

FA1 plan	FA2 location	Evidence to demonstrate
Age group and adjustable goal	Sidebar; age_changed / goal_changed	Change age; preset changes; logs remain
Quick log and containers	Home; log; container_amount	+250 adds 250; half a 1 L bottle adds 500
Live calculations and visuals	current; progress; Home	2,250 / 2,000 gives 112.5%, 0 remaining, full bar
Milestone mascot reactions	milestone; mascot_svg	Set 2,000 ml; reach 1,000, 1,500 and 2,000
Confirmed New Day reset	Home; reset_today	Cancel keeps data; confirm clears today
History and motivation	Progress and Rewards	Real-date streaks; capped points; exports
Optional reminders	reminder_panel on Home	On-screen only; quiet hours and snooze

FA1 RUBRIC COVERAGE

Research and problem understanding 5 marks: user barriers, age-aware goals and annotated design decisions.
App flow and logic planning 10 marks: event loop, amount validation, formulas, reset, date rollover and milestone priority.
Visual storyboard 5 marks: settings, Home states, logging, confirmation, Progress, Rewards and reminder behavior.

Research notes and references

Primary sources support the health context and the choice of deployment platform

HEALTH CONTEXT AND DESIGN IMPLICATIONS

Water supports normal body function and temperature regulation. Dehydration can affect thinking and mood. Keeping a reusable bottle visible and offering water with meals are practical habit cues. These findings support a visible log button and short tips rather than complex health claims. [2]

EFSA total-water reference values include food and all drinks: ages 4-8, 1,600 ml; ages 9-13, 1,900 ml for girls and 2,100 ml for boys; ages 14 and over, including older adults, 2,000 ml for females and 2,500 ml for males. The app explains this distinction rather than labelling its classroom presets as health standards. [1]

SOURCES

[1] EFSA Dietary Reference Values summary report

https://www.efsa.europa.eu/sites/default/files/2017_09_DRVs_summary_report.pdf

[2] CDC About Water and Healthier Drinks

https://www.cdc.gov/healthy-weight-growth/water-healthy-drinks/

[3] Streamlit Community Cloud deployment guide

https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy

[4] Streamlit dependency management

https://docs.streamlit.io/deploy/concepts/dependencies

Sources reviewed 28 October 2026. The supplied FA1 and FA2 briefs define the assessment requirements. The original WaterBuddy storyboard supplies the drop mascot and habit-building concept; this revision aligns the plan with the implemented app.